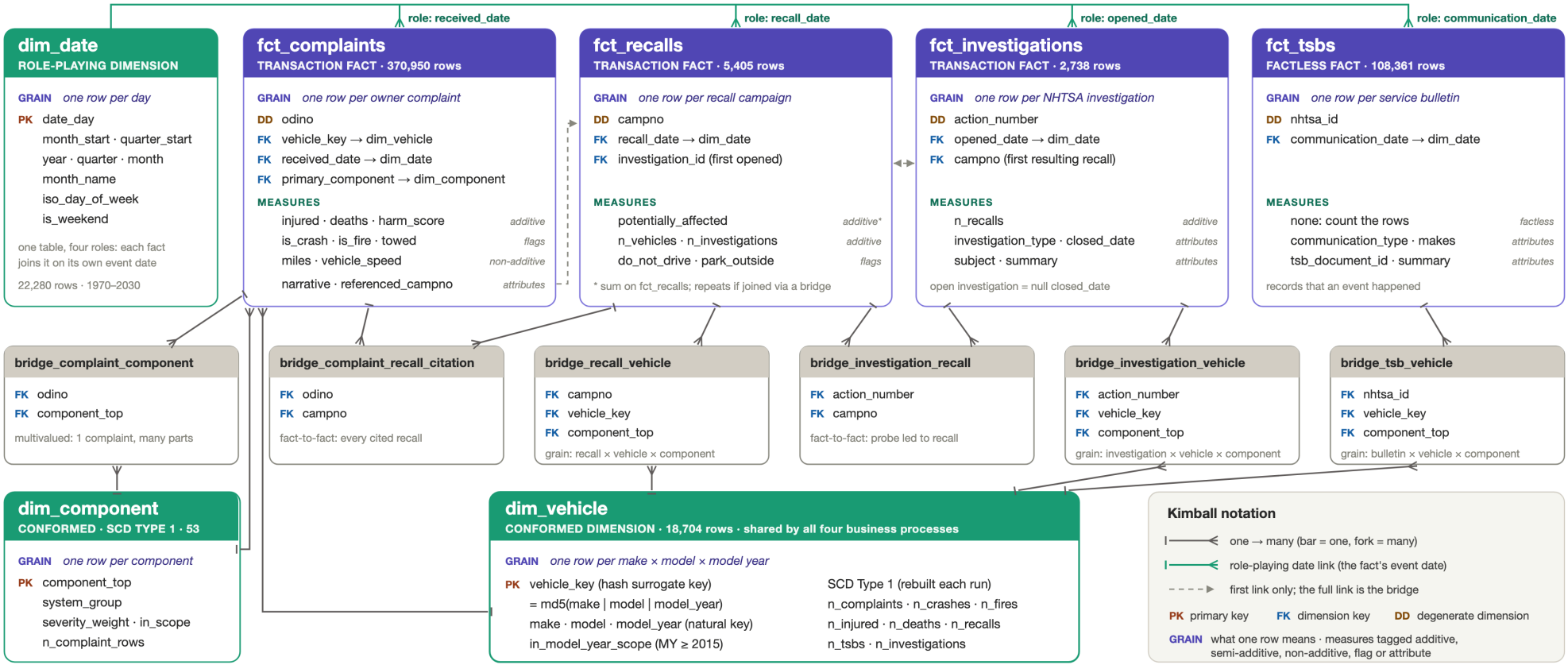
Vroomcheck · Kimball dimensional model

Fact constellation: four business-process fact tables at the atomic grain, sharing conformed dimensions, with bridges for many-to-many links



Four-step design	1 · business processes: complaint, recall, investigation, bulletin → 2 · grain: one row per event → 3 · dimensions: vehicle, date, component → 4 · facts: harm counts, units, counts
Design choices	conformed vehicle key lets every process be compared for the same car (drill-across) · bridges resolve many-to-many links · hash surrogate key instead of integers (deterministic, same on DuckDB and Snowflake) · all dimensions SCD Type 1, history via snapshots is next

The slide these notes go with: Vroomcheck Kimball dimensional model (docs/Vroomcheck_Kimball_Data_Model.png).

Kimball data model · presentation notes

Vroomcheck · Part 1 is the detailed explanation with likely questions · Part 2 is a script to read in 4–5 minutes

Part 1 · Detailed explanation

Follows the diagram: the idea, the facts, the dimensions, the bridges, then the design choices

Opening

Point at the title

- This is the gold layer of our warehouse, designed with **Kimball's dimensional modeling**.
- Kimball designs a warehouse around **business events** rather than entities. Each event gets a **fact table**, and the context you describe it with (which car, which day, which part) goes into **dimension tables**.
- We have four events sharing the same dimensions. Kimball calls this a **fact constellation**: several star schemas joined through conformed dimensions.

1 · The four-step design

Point at the "Four-step design" line at the bottom

- **Step 1, choose the business processes:** an owner files a complaint, a manufacturer issues a recall, NHTSA opens an investigation, a manufacturer issues a service bulletin.
- **Step 2, declare the grain: one row per event**, the most detailed level available (the "atomic grain"). Totals are built on top, never in place of the detail.
- **Step 3, choose the dimensions:** vehicle, date and component.
- **Step 4, choose the facts:** harm counts, affected units and counts.

2 · The four fact tables

Point at the purple boxes along the top

Fact table	Type	Grain (one row per…)	Rows
`fct_complaints`	Transaction fact	owner complaint	370,950
`fct_recalls`	Transaction fact	recall campaign	5,405
`fct_investigations`	Transaction fact	NHTSA investigation	2,738
`fct_tsbs`	Factless fact	service bulletin	108,361

- **Transaction fact:** one row each time the event happens.
- **Factless fact:** `fct_tsbs` has no numeric measures. It just records that a bulletin was issued, so you analyse it by counting rows.
- Each fact has a **GRAIN** line on the diagram, because in Kimball you declare the grain before anything else.

3 · Measures and how they add up

Point at the MEASURES section of fct_complaints

- **Additive** (can be summed across any dimension): `injured`, `deaths`, `harm_score`, `n_recalls`, `potentially_affected`.
- `harm_score` is our own measure: 1 + 3 if crash + 3 if fire + 2 if injuries + 5 if deaths + 1 if towed, so it ranges from 1 to 15. We use it to rank how serious a cluster of complaints is.
- **Flags** (`is_crash`, `is_fire`, `towed`) are counted, e.g. "how many complaints mention a fire".
- **Non-additive** (`miles`, `vehicle_speed`): summing them is meaningless, so you average them instead.
- **The fan-out warning:** `potentially_affected` is additive on `fct_recalls`, but if you join through the vehicle bridge first, it repeats once per vehicle. That's the note on the diagram.

4 · Degenerate dimensions

Point at the orange "DD" labels

- Each fact's own ID, `odino`, `campno`, `action_number` and `nhtsa_id`, sits in the fact table with no dimension table behind it.
- Kimball calls this a **degenerate dimension**. It's like an invoice number: useful for grouping and tracing back to NHTSA, but it has no attributes of its own.

5 · dim_vehicle: the conformed dimension

Point at the big green box at the bottom

- One row per **make × model × model year**: 18,704 vehicles.
- It's **conformed**, meaning **all four business processes use the same vehicle key**. That's what lets us compare complaints, recalls, investigations and bulletins for exactly the same car. Kimball calls this **drill-across**.
- The key is a **hash surrogate key**: `md5(make | model | model_year)`. It's deterministic, so it's the same on every rebuild and on every engine, with no sequence to manage.
- It also carries summary counts (complaints, crashes, fires, recalls…) so you can filter vehicles quickly.

6 · dim_date and dim_component

Point at the green box top-left, then bottom-left

- `dim_date` is a **role-playing dimension**: one calendar table, 1970–2030, used four ways. Each fact joins it on its own event date: received, recall, opened or communication date. Those are the green lines along the top.
- It gives every fact the same calendar attributes: month, quarter, year, weekday, weekend.
- `dim_component` has one row per vehicle component (53), with its system group and a **severity weight** from our ontology seed. Complaints link to it through `primary_component`, and the bridges link to it through `component_top`.

7 · Bridge tables

Point at the grey boxes in the middle row

- Real NHTSA data is **many-to-many**: one recall covers many vehicles, and one complaint can name several components. A plain star schema can't hold that without repeating fact rows.
- Kimball's answer is a **bridge table**. We have six:
  - **Multivalued:** `bridge_complaint_component` (one complaint, many components).
  - **Vehicle bridges:** recall, investigation and bulletin to vehicle, each at the grain fact × vehicle × component.
  - **Fact-to-fact:** `bridge_complaint_recall_citation` (complaints that cite a recall number) and `bridge_investigation_recall` (investigations that led to a recall).
- The dashed arrows are "first link only" shortcut columns, like the first investigation opened for a recall. The full link is always the bridge.

8 · Slowly changing dimensions

- All dimensions are **SCD Type 1**: they're overwritten on each rebuild, so they show the current values and keep no history.
- That fits our data, because a car's make, model and year never change. If we ever need history, dbt snapshots would give us Type 2.

Example query: drill-across with the conformed key

Complaints and recalls for the same car, joined only through `vehicle_key`. Each process is aggregated first, then joined, so nothing is double-counted:

```
with v as (select * from marts.dim_vehicle
           where make = 'HONDA' and model = 'CIVIC' and model_year = 2019),
     c as (select vehicle_key, count(*) as n_complaints, sum(deaths) as deaths
           from marts.fct_complaints group by 1),
     r as (select vehicle_key, count(distinct campno) as n_recalls
           from marts.bridge_recall_vehicle group by 1)
select v.make, v.model, v.model_year, c.n_complaints, c.deaths, r.n_recalls
from v left join c using (vehicle_key) left join r using (vehicle_key);
```

Result: HONDA · CIVIC · 2019 · 426 complaints · 0 deaths · 4 recalls.

Closing line

> *“We designed the gold layer the Kimball way: four business processes at the atomic grain, conformed vehicle, date and component dimensions so every process can be compared for the same car, and bridge tables for the many-to-many links that real NHTSA data has.”*

Questions you might get

Why Kimball and not a normalized (3NF) model?
Our workload is analysis, not transactions. A star schema answers questions with one or two joins from a fact to its dimensions, and people can understand it. A normalized model would need five or more joins for the same question.

Why a hash key instead of an integer surrogate key?
An md5 of make, model and year gives the same key on every rebuild and in every engine, without a sequence or a lookup step. The trade-off is a slightly larger key, which doesn't matter at our size.

What does "conformed" mean?
The same dimension, with the same keys and meanings, is shared by several fact tables. Here dim_vehicle is shared by all four, so a 2019 Honda Civic means the same row everywhere.

Why is fct_tsbs a factless fact?
A bulletin has no natural number to add up. The fact is that it happened, for that vehicle, on that date, so we count rows.

Can you sum potentially_affected?
Yes, on fct_recalls. Just don't sum it after joining through a vehicle bridge, because it repeats once per vehicle row.

Why SCD Type 1?
Vehicle and component attributes don't change over time, so history adds no value. Type 2 through dbt snapshots is the upgrade path if it's ever needed.

Where are the monthly totals?
In `mart_monthly_signal`, an aggregate fact table built from fct_complaints for our spike detection. It isn't on this diagram because it's derived from the atomic facts shown here.

Part 2 · Speaking script

About 680 words, roughly 4.5–5 minutes at a comfortable pace

Section	Time
Introduction	0:30
The four-step design	0:40
The fact tables	0:45
Measures	0:35
The conformed vehicle dimension	0:45
Date and component dimensions	0:30
Bridge tables	0:35
Keys, degenerate dimensions and SCD	0:25
Conclusion	0:20
Total	**≈ 4:45**

Introduction 0:30

Point at the title

This is the data model for our gold layer, designed with Kimball's dimensional modeling. Instead of organising data around entities, Kimball organises it around business events. Each event gets a fact table, and the context we use to describe it, such as which car, which day and which part, goes into dimension tables. We have four events that share the same dimensions, which Kimball calls a fact constellation.

The four-step design 0:40

Point at the bottom band

We followed Kimball's four-step process. First, we chose the business processes: an owner files a complaint, a manufacturer issues a recall, NHTSA opens an investigation, and a manufacturer issues a service bulletin. Second, we declared the grain as one row per event, which is the most detailed level available. Third, we chose the dimensions: vehicle, date and component. And fourth, we chose the facts, such as harm counts and the number of affected vehicles.

The fact tables 0:45

Point at the purple boxes along the top

Along the top are our four fact tables. Complaints, recalls and investigations are transaction facts, meaning one row each time the event happens. We have about 371,000 complaints, 5,400 recalls and 2,700 investigations. Service bulletins are different: they are a factless fact table. A bulletin has no number to add up, so the fact is simply that it happened, and we analyse it by counting rows. Each box also shows its grain, because in Kimball the grain is decided before anything else.

Measures 0:35

Point at the MEASURES section of fct_complaints

We label each measure by how it can be added up. Injuries, deaths and our harm score are additive, so they can be summed in any direction. Flags such as crash and fire are counted. Miles and vehicle speed are non-additive, so we average them instead of summing them. The harm score is our own measure, from one to fifteen, which weights crashes, fires, injuries and deaths to rank how serious a problem is.

The conformed vehicle dimension 0:45

Point at the big green box at the bottom

The most important table is the vehicle dimension, with one row for every make, model and model year, about 18,700 vehicles. It is a conformed dimension, which means all four business processes use exactly the same vehicle key. That is what lets us compare complaints, recalls and investigations for the same car, which Kimball calls drilling across. For example, for a 2019 Honda Civic we can see 426 complaints and 4 recalls with a single join on the vehicle key. The key itself is a hash of make, model and year, so it is identical on every rebuild.

Date and component dimensions 0:30

Point at dim_date, then dim_component

The date dimension is a role-playing dimension. It is one calendar table that each fact uses in its own role: the date a complaint was received, the date of a recall, the date an investigation opened, or the date a bulletin was issued. The component dimension lists the 53 vehicle components, each with a system group and a severity weight from our business rules.

Bridge tables 0:35

Point at the grey boxes in the middle row

Real NHTSA data is many-to-many. One recall covers many vehicles, and one complaint can mention several components. A plain star schema cannot hold that without repeating rows, so we use Kimball's bridge tables. We have six of them: one for complaint components, three that link recalls, investigations and bulletins to vehicles, and two that link facts to each other, such as complaints that cite a recall number.

Keys, degenerate dimensions and SCD 0:25

Point at the DD labels and the notation box

Each fact keeps its own NHTSA ID, such as the complaint number, as a degenerate dimension, so every row can be traced back to the source. All our dimensions are slowly changing dimension Type 1, because a car's make, model and year never change.

Conclusion 0:20

To summarise, our gold layer is a Kimball fact constellation: four business processes at the atomic grain, conformed dimensions so every process can be compared for the same car, and bridge tables for the many-to-many links in real data. Thank you.

Timing tips

- **Running short of time:** drop the harm score sentence in Measures and the SCD sentence. That saves about 20 seconds.
- **Extra time:** show the drill-across query from Part 1 and read out the 2019 Honda Civic result. That covers the "show code" requirement.